Multiple-choice questions

|!|BCEMA0270000001|!| ★★

A DNA template strand with any of the following triplet codes can code for the amino acid arginine:

GCA GCC GCG GCT TCC TCT

Which of the following is an anticodon on the tRNA molecule that carries arginine?

A. CGT
B. CGU
C. GCT
D. UCC

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0270000002|!|

The genetic code is referred to as degenerate because

A. it is the same in every species.
B. some codons do not code for amino acids.
C. a codon may code for more than one amino acid.
D. some codons may code for the same amino acid.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0270000003|!| ★

Many mutations have no effect on the organism because

(1) they are repaired before protein synthesis occurs.
(2) some mutations may produce a different codon for the same amino acid.
(3) transcription of the mutated gene would cease when mutation occurs.

A. (1) only
B. (2) only
C. (3) only
D. (1) and (2) only

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##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0270000004|!|

What causes Down syndrome?

A. independent assortment of chromosomes in the formation of gametes
B. non-disjunction in the formation of gametes
C. crossing over in the formation of gametes
D. gene mutation in the formation of gametes

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0270000005|!|

Which of the following provides evidence for the universal nature of the genetic code?

A. Nucleic acids contain the same bases in all species.
B. In a DNA molecule, the amount of adenine is equal to that of thymine and the amount of guanine is equal to that of cytosine.
C. Uracil replaces thymine in RNA.
D. The same mRNA codon codes for the same amino acid in all species.

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0270000006|!|

Which of the following type(s) of gene mutation can cause a shift in the reading frame of DNA?

(1) deletion
(2) substitution
(3) insertion

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0270000007|!|

Which of the following is the cause of sickle-cell anaemia?

A. an iron-deficient diet
B. infection following a bite from a mosquito
C. a change in the base sequence of a haemoglobin gene
D. a change in the structure of a chromosome

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0270000008|!|

Arrange the following stages of transcription in the correct order.

(1) The mRNA detaches from the DNA template strand.
(2) Free RNA nucleotides bind with complementary bases on the DNA template strand.
(3) mRNA polymerase binds to the DNA at the start of a gene
(4) The two DNA strands of the double helix unwind.

A. (2) → (3) → (1) → (4)
B. (2) → (3) → (4) → (1)
C. (3) → (2) → (1) → (4)
D. (3) → (4) → (2) → (1)

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##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0270000009|!| ★

If a human sperm with 23 autosomes and an X chromosome fertilizes a normal ovum, what would be the chromosome number and sex of the resulting zygote?

	Chromosome number	***Sex***
A.	46	male
B.	46	female
C.	47	male
D.	47	female

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0270000010|!|

Which of the following chemicals can cause changes in DNA?

(1) nitrous acid
(2) nicotine
(3) dioxin

A. (3) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

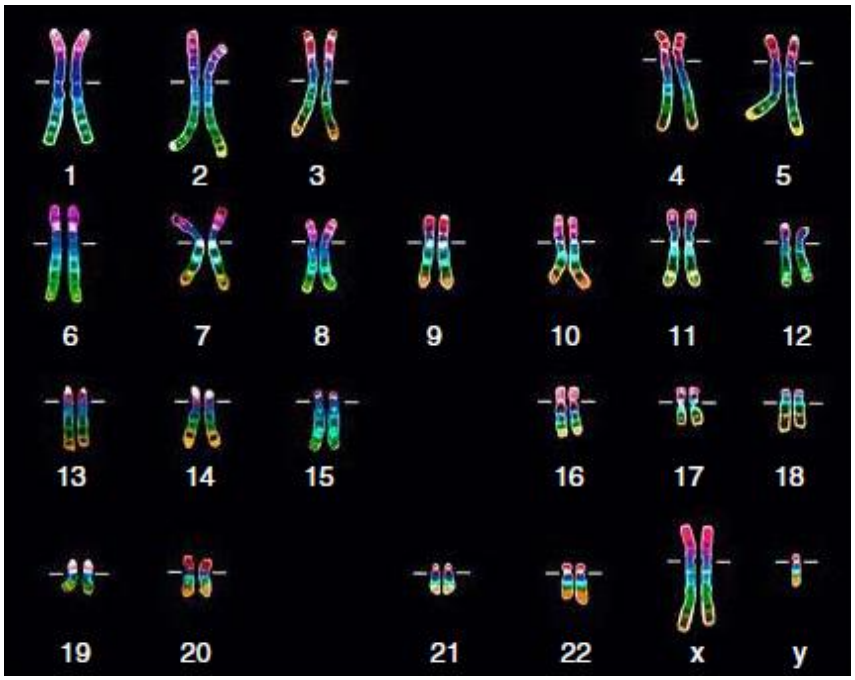
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0270000011|!|

Directions: The follow two questions refer to the photograph below, which shows the karyotype of a person with a chromosomal abnormality.



The number of autosomes in this karyotype is

A. 22.

B. 23.

C. 44.

D. 46.

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0270000012|!| ★

Which of the following may have caused the abnormality?

(1) The paired X chromosomes failed to separate in the formation of the mother's ovum.
(2) The X chromosome and Y chromosome failed to separate in the formation of the father's sperm.
(3) Two sperms fused with an ovum simultaneously.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0270000013|!| ★

A length of double-stranded DNA contains 360 nucleotides and codes for polypeptide X. What is the maximum number of amino acids that polypeptide X can contain?

A. 60
B. 120
C. 180
D. 360

##A##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMB0270000014|!| ★

The following shows the sequence of bases on part of the coding strand of a gene.

ATACGGAACTACATG

An mRNA molecule is produced during transcription of the gene. Which of the following is the sequence of the bases on the mRNA which corresponds to the sequence shown above?

A. TATGCCTTGATGTAC
B. UAUGCCUUGAUGUAC
C. AUACGGAACUACAUG
D. CGCATTCCAGCACGT

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0270000015|!| ★★

A polypeptide contains the following amino acid sequence:

Glu – Asn – Thr – Pro

The table shows the anticodon on the tRNA molecules that carry these amino acids.

Amino acid	tRNA anticodons
Asn	UUA, UUG
Glu	CUU, CUC
Pro	GGA, GGG, GGU, GGC
Thr	UGA, UGG, UGU, UGC

Which of the following is the base sequence on the coding strand of DNA that codes for the above amino acid sequence?

A. GAAAAUACCCCU
B. GAAAATACCCCT
C. CUUUUAUGGGGA
D. CTTTTATGGGGA

##B##

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Skill assessed	Mark
Analysing data	1

|!|BCEMB0270000016|!| ★

A polypeptide contains the following amino acid sequence:

Gly – Leu – Lys – Val

The table shows the mRNA codons that code for these amino acids.

Amino acid	mRNA codons
Gly	GGU, GGC, GGA, GGG
Leu	CUU, CUC, CUA, CUG
Lys	AAA, AAG
Val	GUU, GUC, GUA, GUG

Which of the following is the base sequence of the coding strand of DNA that codes for the above amino acid sequence?

A. CCCGAATTTCAA
B. CCCGAAUUUCAA
C. GGGCTTAAAGTT
D. GGGCUUAAAGUU

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0270000017|!| ★

The table below shows the percentages of the different nitrogenous bases found in two samples of nucleic acids.

	Base				
Sample	**Uracil**	**P**	**Q**	**Cytosine**	**R**
1	20%	30%	25%	25%	0%
2	0%	26%	24%	26%	24%

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Which of the following combinations correctly identifies bases P, Q and R?

	P	***Q***	***R***
A.	adenine	guanine	thymine
B.	guanine	thymine	adenine
C.	guanine	adenine	thymine
D.	thymine	guanine	adenine

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0270000018|!| ★

Directions: The following two questions refer to the two tables below. Table 1 shows the sequence of amino acids in a certain part of a normal polypeptide and an abnormal polypeptide. Table 2 shows the mRNA codons that can code for these amino acids.

Table 1

	Amino acid sequence
Normal polypeptide	His – Thr – Ser – Gly
Abnormal polypeptide	His – Ser – Ser – Gly

Table 2

Amino acid	**mRNA codons**
Gly	GGU, GGC, GGA, GGG
His	CAU, CAC
Ser	AGU, AGC
Thr	ACU, ACC, ACA, ACG

Which of the following changes in the coding strand of the DNA is/are likely to cause the formation of the abnormal polypeptide?

(1) A base C is substituted with a base G
(2) A base G is added
(3) The 5th and the 6th bases are inverted.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

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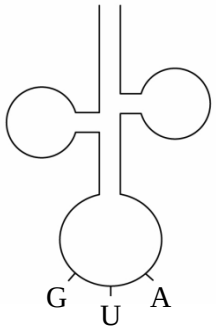
##C##

Skill assessed	Mark
Analysing data	1

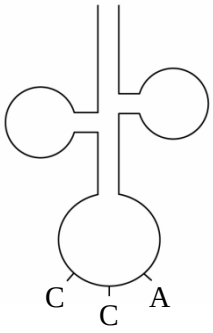
|!|BCEMB0270000019|!| ★

Which of the following tRNA molecules is ***not*** involved in the formation of the amino acid sequence of the abnormal polypeptide?

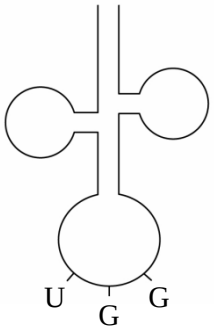
A.



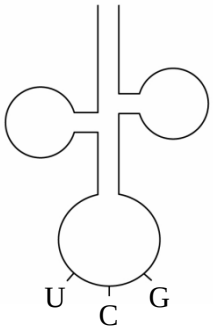
B.



C.



D.



##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0270000020|!| ★

A polypeptide contains the following amino acid sequence:

alanine – leucine – tyrosine – glutamic acid

The table below gives the base sequences on the template strand of DNA that specify these amino acids.

Alanine	Leucine	Tyrosine	Glutamic acid
CGT	GAC	ATA	CTC

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Which of the following shows the anticodons on the tRNA molecules needed for the synthesis of this amino acid sequence?

A. CGT, GAC, ATA, CTC
B. CGU, GAC, AUA, CUC
C. GCA, CTG, TAT, GAG
D. GCA, CUG, UAU, GAG

□

##B##

Skill assessed	Mark
Applying concepts	1

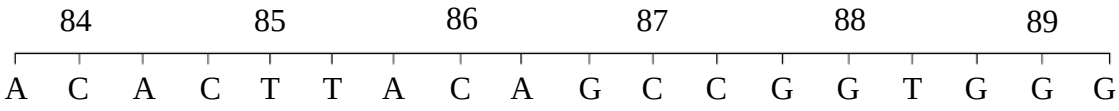
|!|BCEMB0270000021|!| ★

Directions: The following three questions refer to the table below which shows the 64 mRNA codons and the amino acids or signals they specify.

First base	Second base				Third base
	U	C	A	G	
U	Phenylalanine	Serine	Tyrosine	Cysteine	U
	Phenylalanine	Serine	Tyrosine	Cysteine	C
	Leucine	Serine	(Stop)	(Stop)	A
	Leucine	Serine	(Stop)	Tryptophan	G
C	Leucine	Proline	Histidine	Arginine	U
	Leucine	Proline	Histidine	Arginine	C
	Leucine	Proline	Glutamine	Arginine	A
	Leucine	Proline	Glutamine	Arginine	G
A	Isoleucine	Threonine	Asparagine	Serine	U
	Isoleucine	Threonine	Asparagine	Serine	C
	Isoleucine	Threonine	Lysine	Arginine	A
	Methionine (Start)	Threonine	Lysine	Arginine	G
G	Valine	Alanine	Aspartic acid	Glycine	U
	Valine	Alanine	Aspartic acid	Glycine	C
	Valine	Alanine	Glutamic acid	Glycine	A
	Valine	Alanine	Glutamic acid	Glycine	G

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The following shows the sequence of bases on part of the template strand of DNA.



What would be the anticodon of the first tRNA molecule to enter a ribosome during translation?

A. AUG
B. ATG
C. UAC
D. TAC

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0270000022|!| ★

Which of the following shows the base sequence of the mRNA transcribed from the DNA sequence shown above?

A. ACACTTACAGCCGGTGGG
B. TGTGAATGTCGGCCACCC
C. UGUGAAUGUCGGCCACCC
D. ACACUUACAGCCGGUGGG

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0270000023|!| ★

Which of following statements best describes what the polypeptide chain would be like if the 90th triplet code is ATC and the 91st triplet code is CTT on the DNA template strand?

A. The polypeptide chain would be no more than 89 amino acids long.
B. The 90th amino acid would be leucine.
C. The polypeptide chain would be 90 amino acids long.
D. The polypeptide chain would be more than 90 amino acids long.

##A##

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Skill assessed	Mark
Applying concepts	1

|!|BCEMC0270000024|!|

Which of the following statements about tRNAs is ***incorrect***?

A. tRNAs are single-stranded.

B. tRNAs are synthesized in the endoplasmic reticulum.

C. There is at least one tRNA for each of the 20 amino acids.

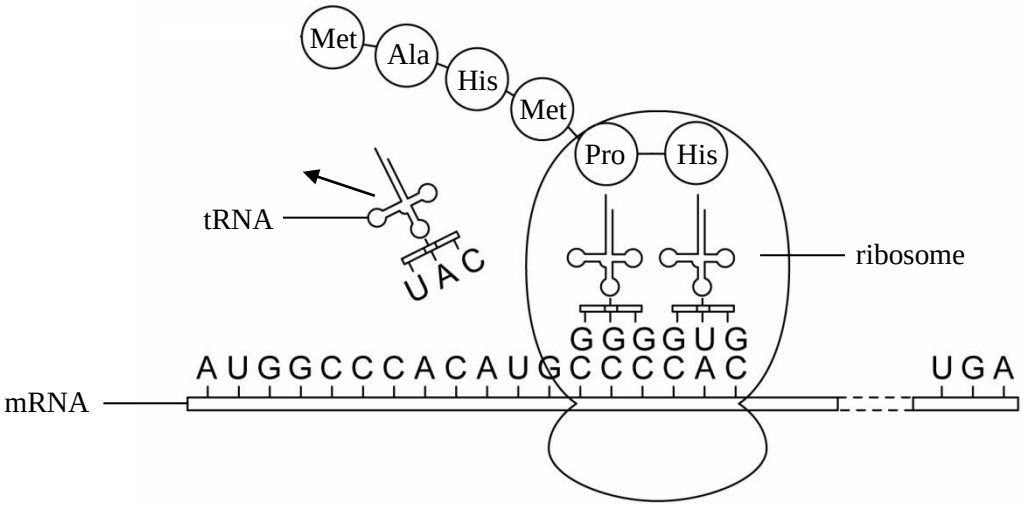
D. tRNAs are required for translation.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0270000025|!| ★

The diagram below shows the process of translation.



Which of the following statements about the translation process shown above is ***incorrect***?

A. The ribosome moves along the mRNA from left to right.

B. A peptide bond is formed between proline and histidine.

C. tRNA molecules can be reused.

D. UGA is the start codon.

##D##

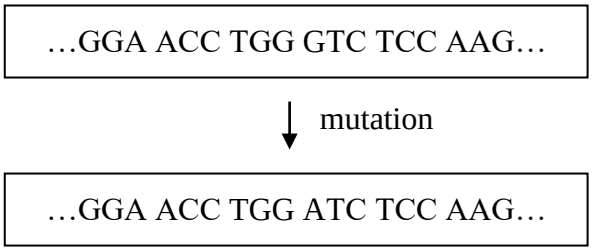
Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0270000026|!|

Directions: The following three questions refer to the information below.

β-thalassaemia is caused by mutations in the gene that codes for the β-chain of haemoglobin. The figure below shows a mutation in the template DNA of the gene.



Which of the following is the normal sequence of the bases on the non-template DNA in this part of the gene?

A. CCT TGG ACC CAG AGG TTC
B. CCT UGG ACC CAG AGG UUC
C. CCT TGG ACC TAG AGG TTC
D. CCU UGG ACC UAG AGG UUC

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0270000027|!|

The type of mutation shown in the figure is

A. deletion.
B. insertion.
C. substitution.
D. inversion.

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0270000028|!| ★

The table below which shows the 64 mRNA codons and the amino acids or signals they specify.

First base	Second base				Third base
	U	C	A	G	
U	Phenylalanine	Serine	Tyrosine	Cysteine	**U**
	Phenylalanine	Serine	Tyrosine	Cysteine	**C**
	Leucine	Serine	(Stop)	(Stop)	**A**
	Leucine	Serine	(Stop)	Tryptophan	**G**
C	Leucine	Proline	Histidine	Arginine	**U**
	Leucine	Proline	Histidine	Arginine	**C**
	Leucine	Proline	Glutamine	Arginine	**A**
	Leucine	Proline	Glutamine	Arginine	**G**
A	Isoleucine	Threonine	Asparagine	Serine	**U**
	Isoleucine	Threonine	Asparagine	Serine	**C**
	Isoleucine	Threonine	Lysine	Arginine	**A**
	Methionine (Start)	Threonine	Lysine	Arginine	**G**
G	Valine	Alanine	Aspartic acid	Glycine	**U**
	Valine	Alanine	Aspartic acid	Glycine	**C**
	Valine	Alanine	Glutamic acid	Glycine	**A**
	Valine	Alanine	Glutamic acid	Glycine	**G**

Which of the following is the sequence of amino acids coded for by the mutant gene shown above?

A. … Glycine – Threonine – Tryptophan – Valine – Serine – Lysine …
B. … Glycine –Threonine – Tryptophan
C. … Proline – Tryptophan – Threonine – Tryptophan – Arginine – Phenylalanine …
D. … Proline – Tryptophan – Threonine

□

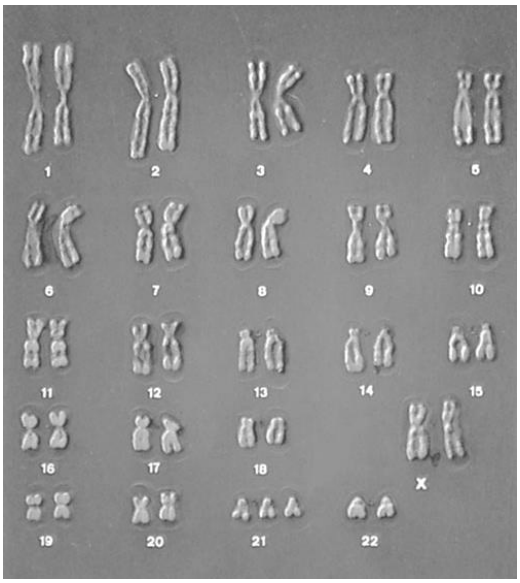
##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0270000029|!| ★

Directions: The following two questions refer to the photograph below. The photograph shows the karyotype of a person, K, who has a genetic abnormality.



Which of the following statements are correct?

(1) K has 47 chromosomes in each of his/her body cell.
(2) K is a female.
(3) All the offspring of K will have this abnormality.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0270000030|!| ★

Which of the following can result in the genetic abnormality?

(1) random alignment of chromosomes during meiosis
(2) non-disjunction of homologous chromosomes to separate during meiosis I
(3) non-disjunction of sister chromatids to separate during meiosis II

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0270000031|!| ★

In a study, three groups of cells are subject to different conditions as shown below.

Group of cells	Conditions
(1)	Ultraviolet radiation
(2)	Sulphur dioxide
(3)	Nitrous acid

It is known that protein X is produced in cells in response to DNA damage. In which groups of cells would large quantities of protein X be found?

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0270000032|!|

RNA does ***not*** contain

A. adenine.
B. cytosine.
C. guanine.
D. thymine.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0270000033|!|

Which of the following comparisons between DNA and RNA is ***incorrect***?

	DNA	***RNA***
A.	is double-stranded	is single-stranded
B.	has the sugar deoxyribose	has the sugar ribose
C.	has the base thymine	has the base uracil
D.	present in the nucleus	absent in the nucleus

##D##

Skill assessed	Mark
Understanding basic concepts	1

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